Novel In-Plane Flexure-Based Clamp Designs

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Introduction

Problem with Traditional Clamps:
- Set screws create point loads and damage rods
- Require flats on rods, complicating manufacturing
- Prone to loosening under vibrations

Why Flexure-Based Clamps?
- Provide uniform clamping force
- Easier to manufacture with wire EDM
- Allow easy rod insertion and alignment
- Traditional designs need radial access, limiting use in tight spaces

Our Contribution:
- Developed planar flexure clamps using tapered screws
- Optimized for manufacturability and performance
- Eliminated requirement for radial access

Design Overview

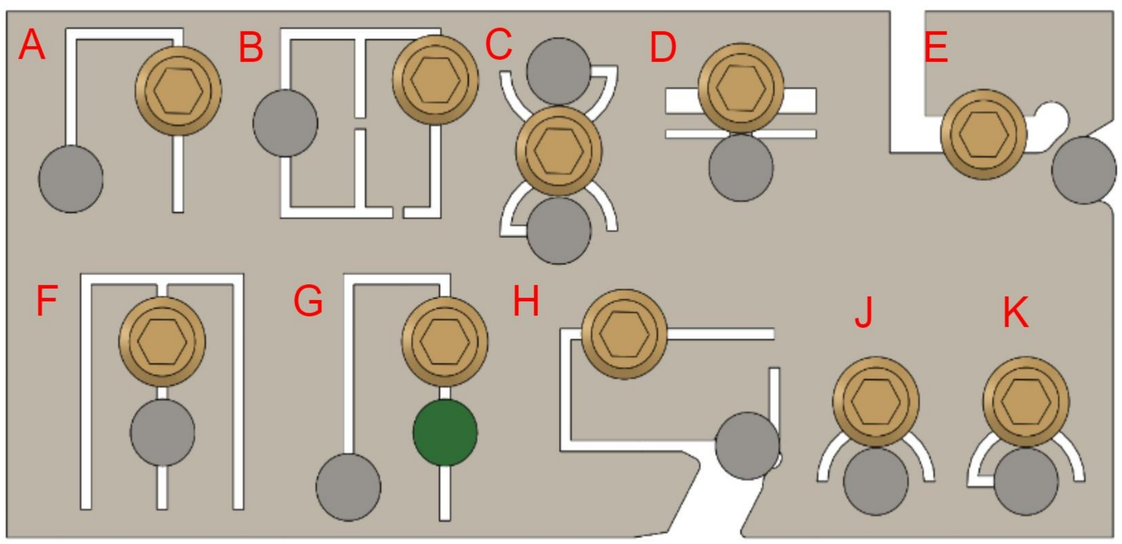
Clamps: Aluminum 6061-T6
Rods: 6 mm Stainless Steel 316
Manufacturing: Wire EDM on Sodick ALN400G iGE+
Surface Roughness:
Clamps: 1801 nm (Ra), Rods: 50 nm

Screw-Tightened Clamps (Designs A, B, C, D, G, J, K):
- Use tapered screws to tighten clamps
- Provide uniform force around rods
- *Design C* clamps two rods at once

Insert-and-Remove Screw Clamps (Designs F, G):
- Screw widens aperture for rod insertion
- Removing screw clamps the rod
- Simplifies use, reduces wear, and prevents creep

Side Entry Flexure Lock Clamps (Designs E, H):
- Rods inserted from the side
- Ideal for confined spaces
- Flexure lock activated by tapered screw

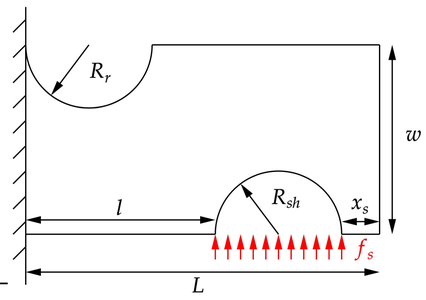
Panel Flexure Lock Clamps (Designs L, M, N):
- Clamp 3 mm Acrylic sheets at 90° to base plate
- Made from 6 mm Delrin via laser cutting
- Evenly distribute clamping forces



Rod Flexure Clamp Designs

Mathematical Modeling

- Predict clamping pressure based on torque
- Modeled as a cantilever beam



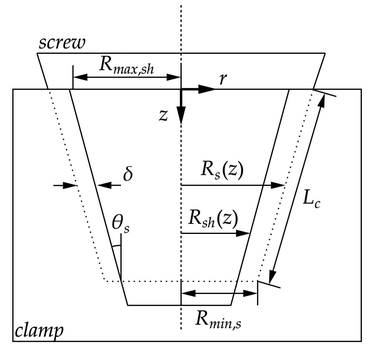
Tapered Screw Mechanics

Interference Pressure

$$p_s = \frac{\delta \cos\theta_s}{R_s(z)\left[\frac{1}{E_c}\left(\frac{r_c^2+R_s(z)^2}{r_o^2-R_s(z)^2}+v_c\right)+\frac{1}{E_s}(1-v_s)\right]}$$

Torque Relation

$$\mathrm{T} = \frac{\mathrm{F_{S,y}}\left[\frac{P}{\pi}+\mu_s\left(d_m-0.649535P\right)\sec\left(\tan^{-1}\left(\tan\alpha\cos\left(\tan^{-1}\left(\frac{l_h}{\pi d_n}\right)\right)\right)\right)\right]}{2\tan\theta_s}$$



Clamp-Screw Geometric Relations:

$$r_o = R_{sh}(z) + x_s$$
$$\delta = R_s(z=0) - R_{sh}(z=0)$$
$$R_s(z) = R_{\min,s} + (L_C\cos\theta - z)\tan\theta_s$$
$$R_{sh}(z) = R_{\max,sh} - z\tan\theta_s$$

Pressure on Rod: $F_{s,y} = \iint p_s dA \rightarrow p_r = \frac{F_{s,y}}{R_r t}$

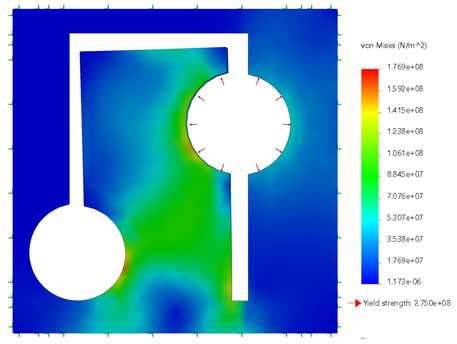
- Direct proportionality between clamping pressure and applied torque

Finite Element Analysis (FEA)

Purpose:
- Validate mathematical model
- Simulate clamp behavior under load

Method:
- Applied radial pressure at screw hole interface
- Analyzed stress distribution

Results:
- FEA results highlight stress concentrations at clamping interface



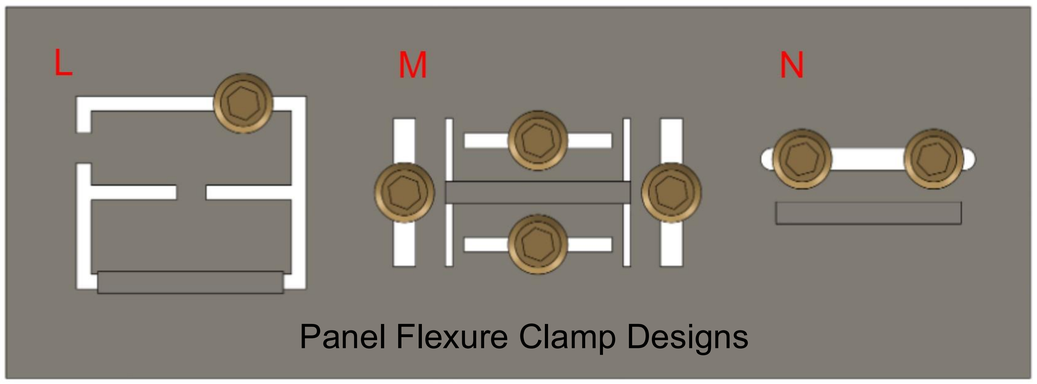
Conclusions

Efficient Design:
- Simple equations predict clamping pressure
- Design allows for easy manufacturability

Versatile Clamps:
- Multiple designs address various needs
- Suitable for confined spaces and different materials

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Panel Flexure Clamp Designs